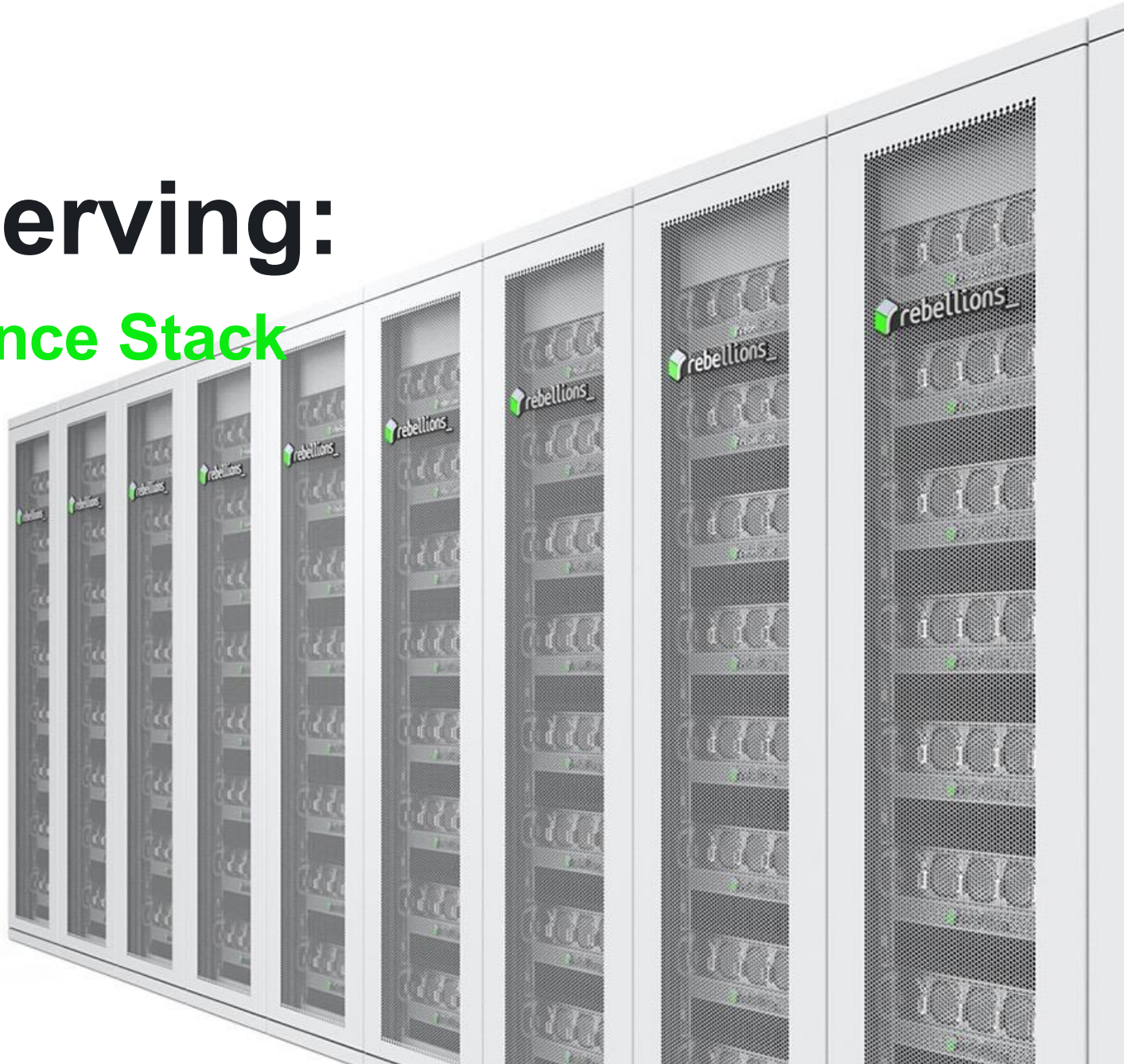
rebellions_

From Silicon to Serving:

Inside the Rebellions Inference Stack



Taejoon Kong

AI Solutions Lead, Rebellions

SCAN FOR LINKS

Docs, white paper, publications, GitHub

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Compute is getting faster. Data must keep up.

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From a PyTorch model to rack-scale LLM serving

03 **How to use RebelServer** p. 29

Validated recipes, live serving and demos



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Rebellions: an AI inference company, from silicon to the rack





Fabless NPU company founded in 2020 in Seoul, built around one workload: AI inference.

Silicon

ATOM · ATOM-Max
Rebel100

Systems

RebelServer · RebelRack
RebelPOD

Software

RBLN SDK: compiler, runtime, profiler
vLLM · PyTorch · Kubernetes

400+

patents across hardware and software

300+

engineers and specialists, 50+ PhDs

6

Locations: Korea, Japan, Saudi Arabia, Singapore, Taiwan, United States and more coming online in 2027.

In production: hundreds of racks deployed for governments and enterprises

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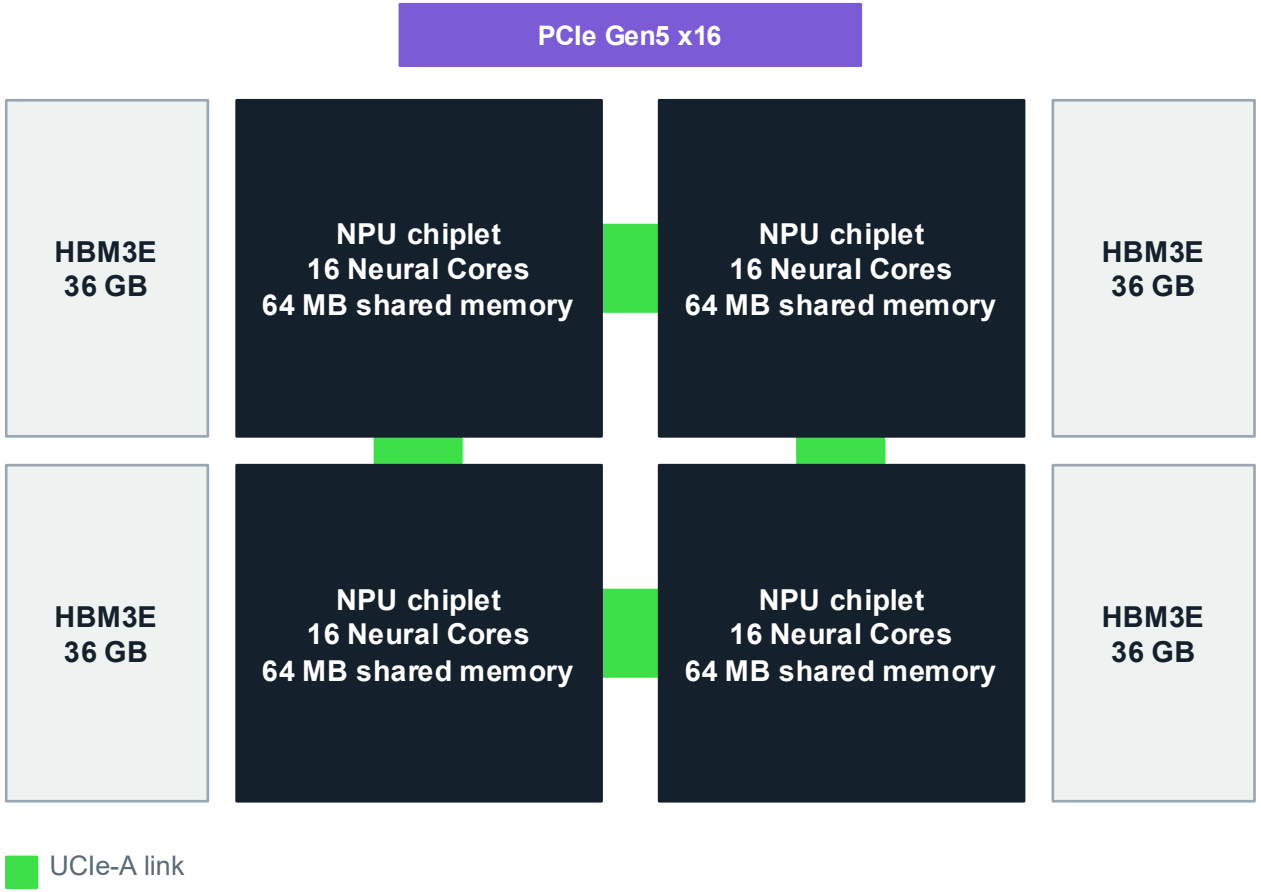
PART 1

Hardware

Compute is getting faster. Data must keep up.

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Rebel100: four chiplets, one accelerator



2,048
TFLOPS FP8

4.8 TB/s
HBM3E · 144 GB

512 MB
on-chip SRAM

600 W
air-cooled card

Built as four, used as one.

Software sees a single NPU.

Sources: C.-H. Yu et al., ISSCC 2026, paper 2.2; Rebel100 product brief v1.6.

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RebelServer Specification

Feature	Description
Form Factor	5U (8x RebelCard™ PCIe Gen 5 cards)
CPU	2x AMD EPYC 9355 (32C 64T, 3.55GHz, 280W) [EPYC 5th Gen]
Memory	1.5TB (24x 64GB DDR5)
Disk	OS: 1.92TB NVMe x 2 Data (KV Cache): 1.92TB NVMe x 2
Network	4x 400G 1-Port QSFP112-DD 1x 100G 2-Port QSFP56 1x 10G/25G 2-Port SFP28 (OCP 3.0)
PSU	6x 2700W
Typical/ Max Power Consumption[3)]	4kW/7kW
Accelerator	8x Rebel™ 100
OS	Red Hat or Ubuntu (plus Rebel Software Stack)



1) Subject to change based on customer requirements
2) SFP28 or RJ45 available depending on customer requirements
3) Theoretical maximum power consumption based on specifications. Actual power consumption will not exceed 7 kW, typically hovering around 4-6kW at most under practical workloads.

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RebelRack Configuration (2-Node Starter)

Feature	Description
1 x NPU Rack	2 x RebelServer nodes, each with 8 x RebelCards™
1 x Management Rack	1 x 800G backend switches 1 x 25G frontend switches 1 x 1G out-of-band switches 3 x management servers (optional) Non-HA networking
42RU data center rack	
PDUs, cables and accessories included	
Air-cooled rack deployment	
Can be upgraded to a 4 Node Standard Configuration (both racks)	

NPU Rack
2 Node



14kW

Management
Rack



< 5kW

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From server to rack

8 × Rebel100

RebelCard

2 PFLOPS FP8

144 GB HBM3E

600 W max



RebelServer

5U · 8 cards

16.4	PFLOPS FP8
1,152 GB	HBM3E
7 kW	max
4 kW	Typical

RebelRack

42U · 4 servers · 32 cards

65.5	PFLOPS FP8
4.6 TB	HBM3E
28 kW	max, air-cooled
28 kW	max, air-cooled



Sources: RebelServer datasheet v1.6; RebelRack standard spec. Peak figures are sums of card specifications.

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RebelPOD Specification (32-Node)

Starting Configuration

32-Node Cluster (8 x RebelRack and 1 x Management Rack)

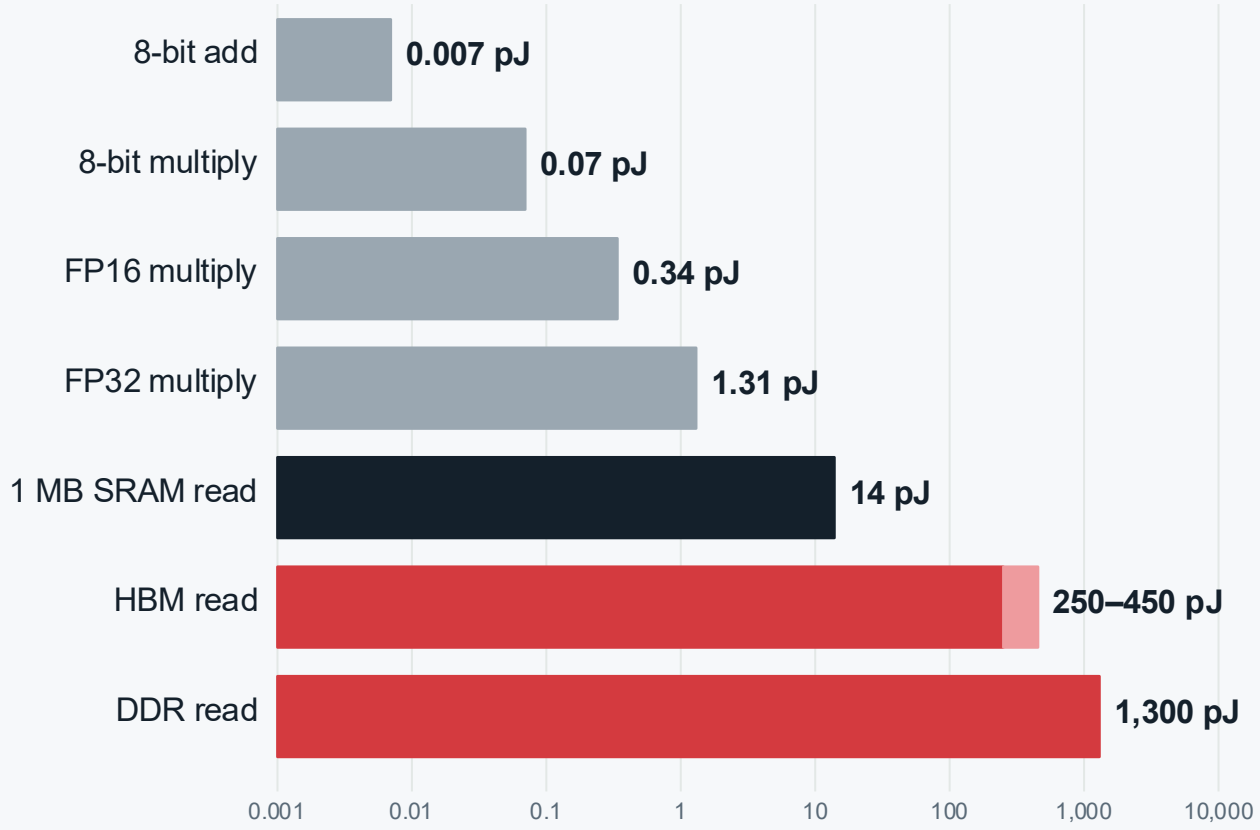


Preset Configurations | 64 nodes (16 x RebelRack) > 128 nodes (32 x RebelRack) > 256 nodes (64x RebelRack) and networking

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Inference is now a data-movement problem

Energy per operation, 7 nm, log scale (pJ)



180–330×

energy to read one FP16 value from HBM, compared with multiplying it

Every two years

3.0× peak compute

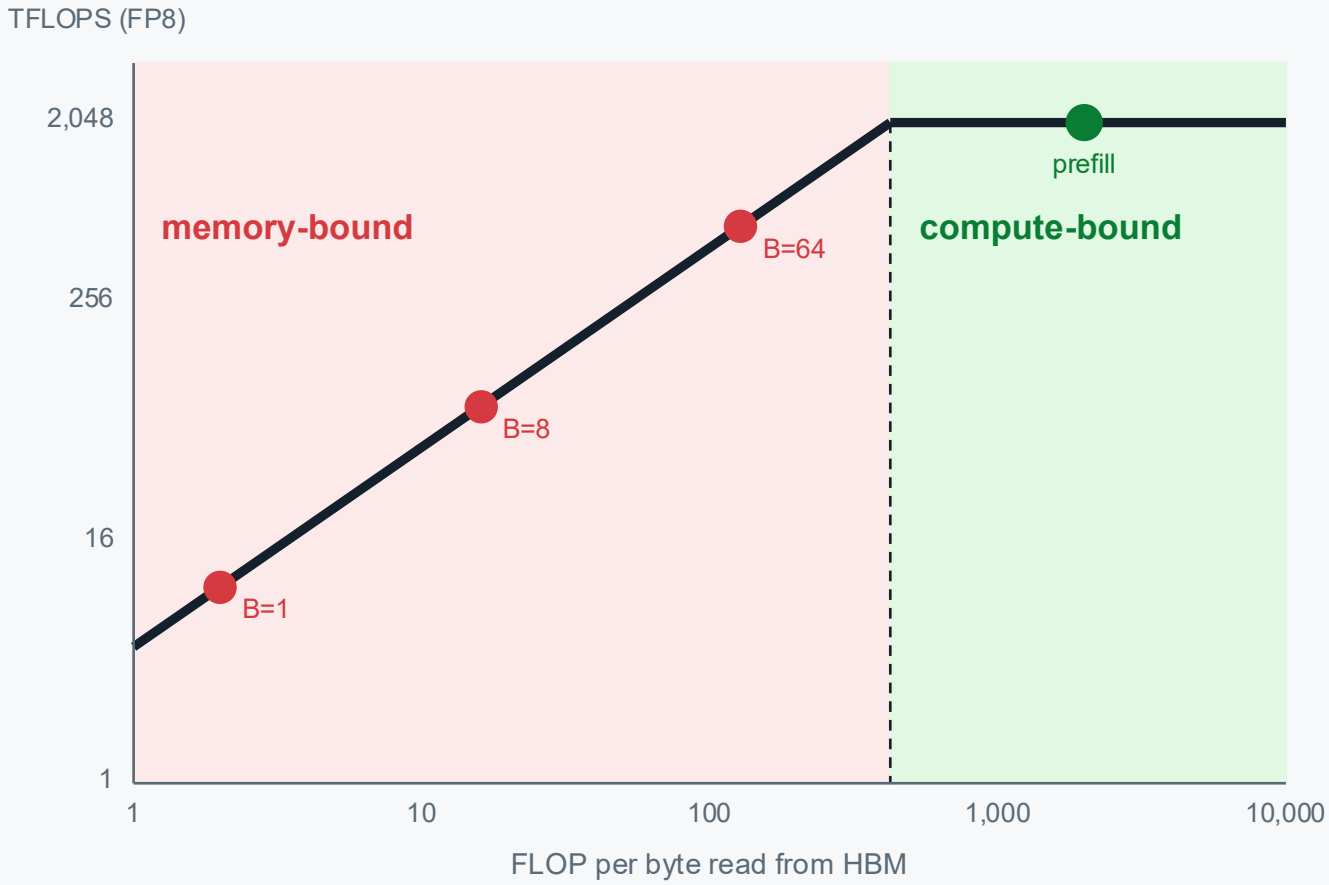
1.6× DRAM bandwidth

1.4× interconnect

Sources: N. P. Jouppi et al., ISCA 2021, Table 2 (7 nm; memory per 64-bit access, I/O energy only); A. Gholami et al., 'AI and Memory Wall', IEEE Micro 2024.

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The roofline is the ideal. Stalls pull real performance below it.



Why real performance falls below the line

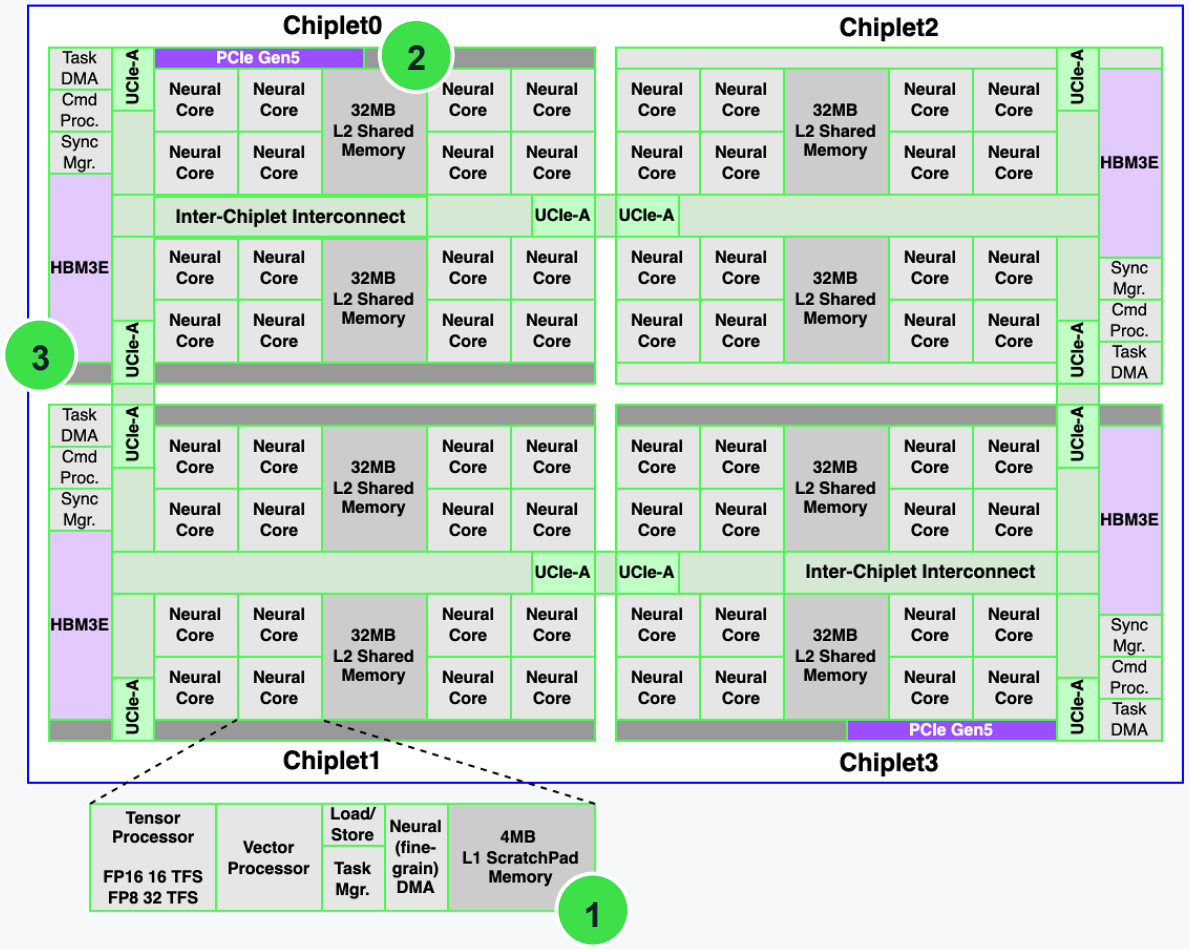
1. Data sits too far away
   Keep data close
2. Data arrives late
   Keep data flowing
3. Units fall out of step
   Keep execution in step

~~How many FLOPS?~~ → **How few stalls per byte?**

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Keep the work inside the core, then the cluster

1 · KEEP DATA CLOSE



Aggregate bandwidth, closest tier first

1 Neural Core scratch pad

4 MB × 64 cores (aggregated)

128 TB/s

2 Cluster shared memory

32 MB × 8 clusters (aggregated)

64 TB/s

3 HBM3E

144 GB, off-chip (aggregated)

4.8 TB/s

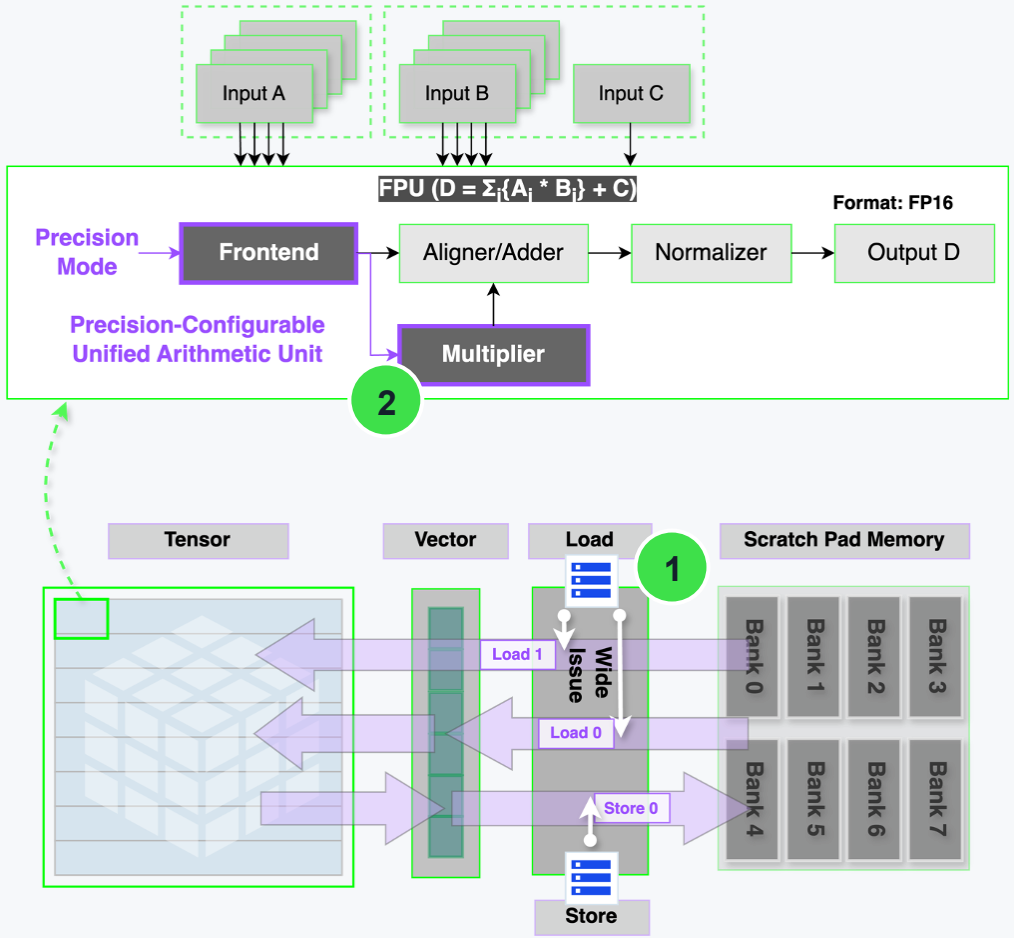
Every byte reused in ① or ② is one less trip to ③.

Sources: Rebellions, Rebel100 white paper (Fig. 1); C.-H. Yu et al., ISSCC 2026, paper 2.2 (aggregate SRAM bandwidth).

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Feed the next operands while computing the current ones

2 · KEEP DATA FLOWING



1 Wide-issue load / store

Feeds to the tensor and vector units and the write-back are issued in the same step.
Eight scratch-pad banks serve them in parallel.

2 One multiplier for every precision

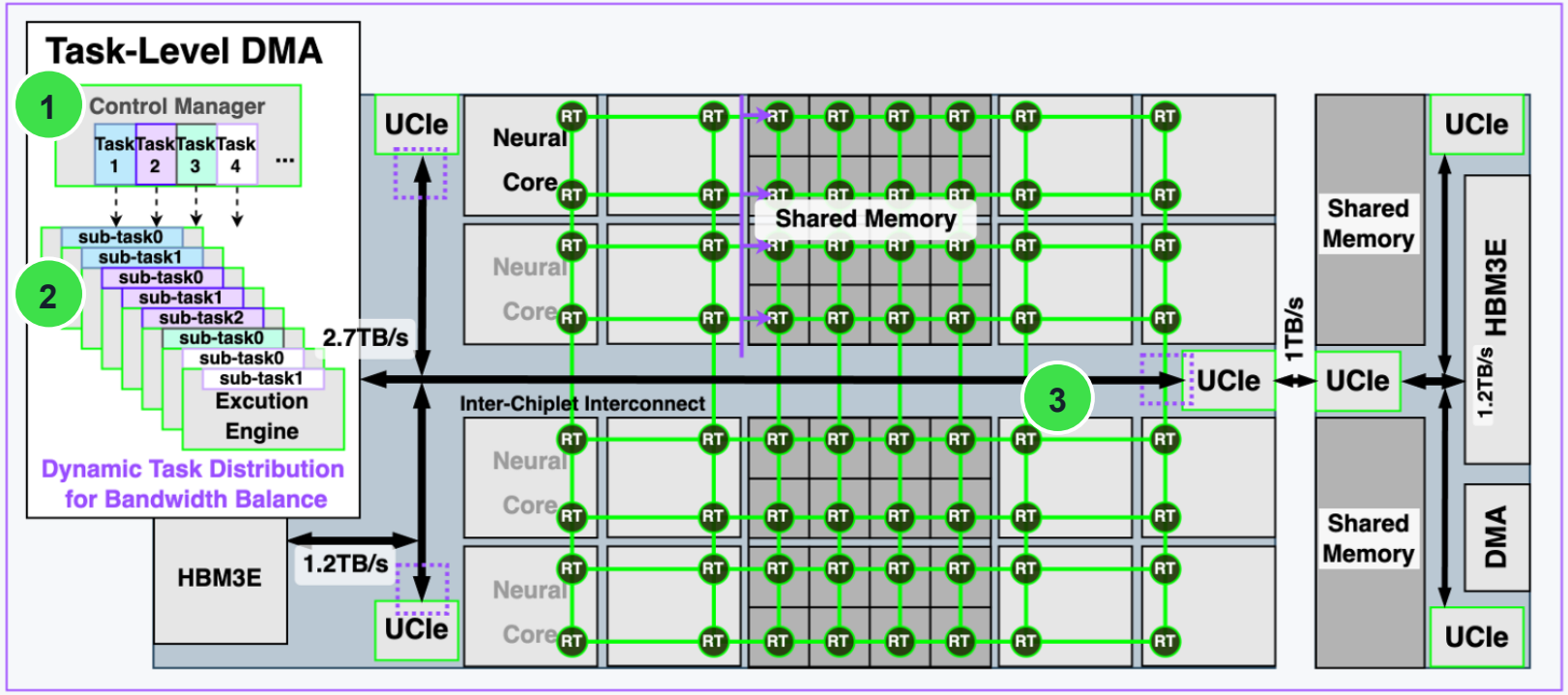
The frontend reads each operand's mode, FP8 or FP16; one shared multiplier runs both.
32 TFLOPS FP8 · 16 TFLOPS FP16 per core.

Sources: Rebellions, Rebel100 white paper (Fig. 2); C.-H. Yu et al., ISSCC 2026, paper 2.2.

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Split every transfer and spread it across the mesh

2 · KEEP DATA FLOWING



One mesh of 256 routers spans all four chiplets: many paths at once, not one shared bus.

1 Split

The control manager breaks each data-movement task into sub-tasks.

2 Spread

Sub-tasks go to 8 execution engines per chiplet, balancing bandwidth.

3 Route

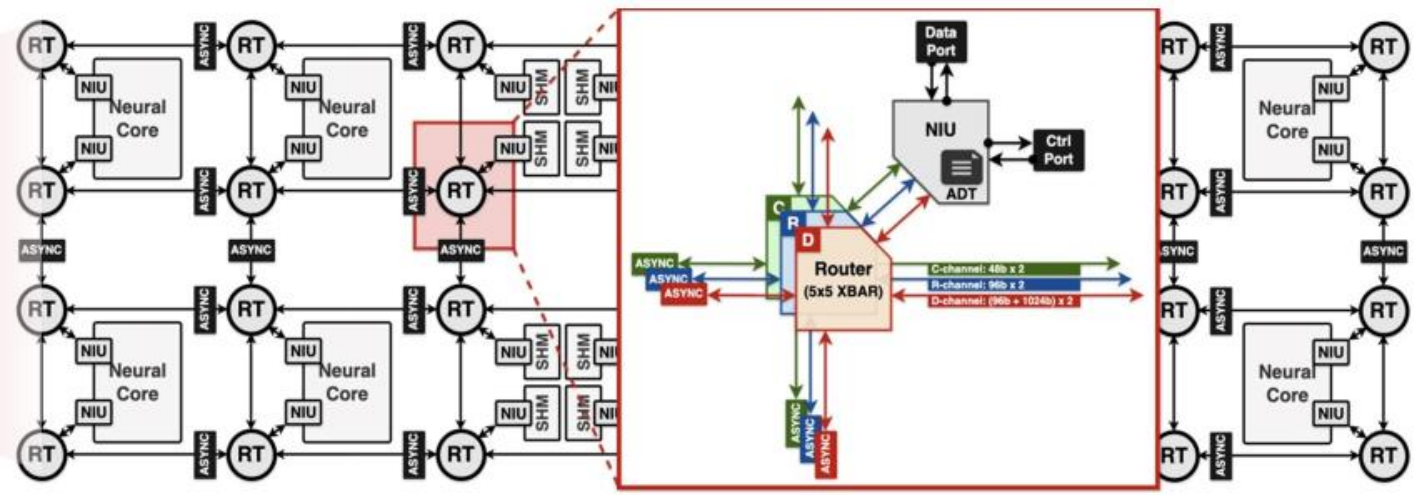
Engines push them over the router mesh, to local and remote HBM3E at once.

Sources: Rebellions, Rebel100 white paper (Fig. 3); C.-H. Yu et al., ISSCC 2026, paper 2.2.

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Keep small messages out of the bulk-data queue

2 · KEEP DATA FLOWING



Mesh routers and one 5×5 crossbar router, zoomed: three channels in every router

D	**Data**		Bulk data, e.g. KV cache
R	**Request**		Requests and responses
C	**Control**		Control messages

Separate lanes

A small request never queues behind a 1,024-bit bulk transfer. It removes waiting; it does not add bandwidth.

QoS on weighted round-robin

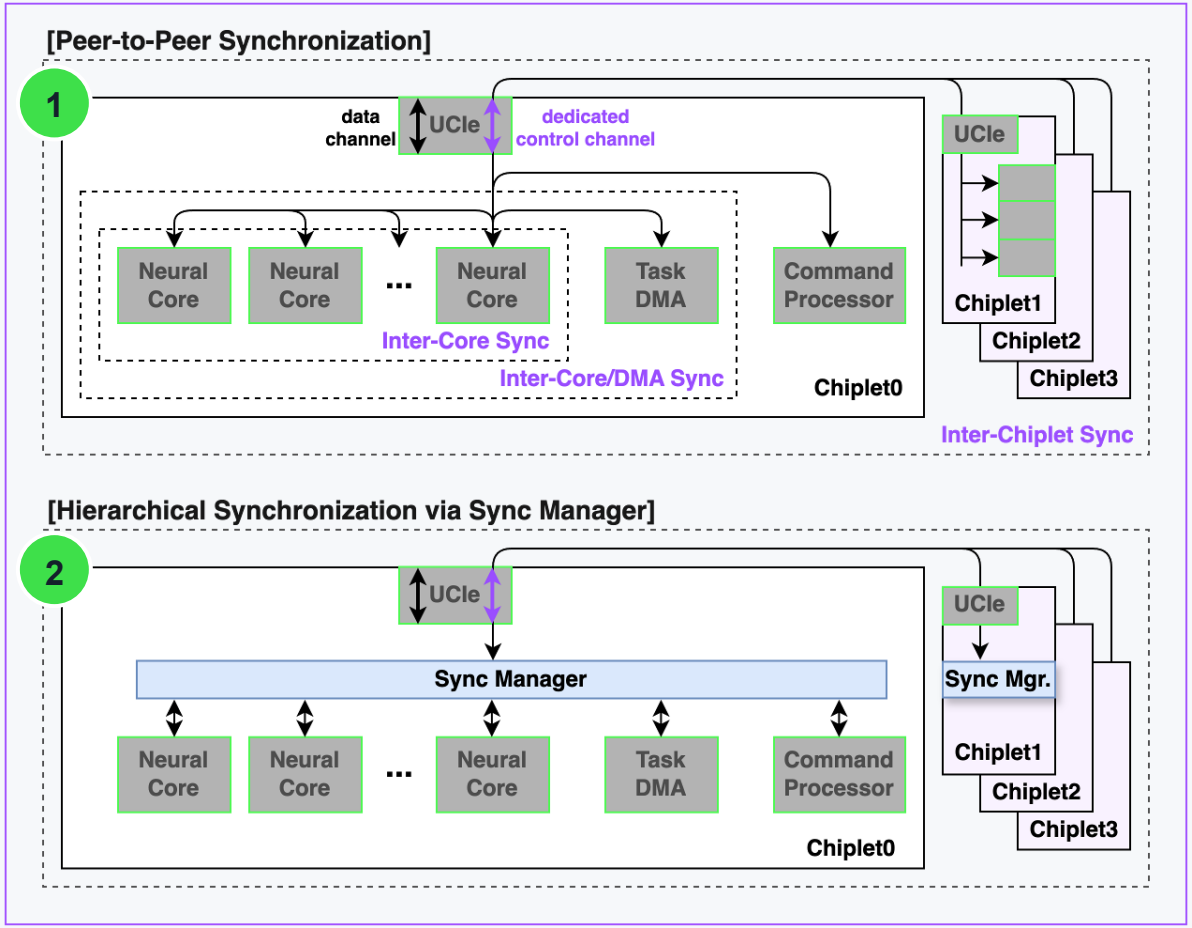
Every router takes turns by weight, retunable at run time. Per-task QoS in the DMA shortens the slowest transfer, and a step ends when its slowest transfer ends.

Source: C.-H. Yu et al., ISSCC 2026, paper 2.2 (Fig. 2.2.2, mesh and router).

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Two hardware sync modes on one dedicated channel

3 · KEEP EXECUTION IN STEP



1 Peer-to-peer sync

The producer signals its consumer directly.
Best for one-to-one handoffs: a DMA tile to the core that uses it.

2 Sync Manager

Units report to their chiplet's hub and are released together.
Best for lock-step across many cores and chiplets.

Both on a dedicated sync channel, never queued behind data

Sources: Rebellions, Rebel100 white paper (Fig. 4); C.-H. Yu et al., ISSCC 2026, paper 2.2.

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Where Rebel100 saves energy: shorter trips, fewer waits

E_{token} ≈ energy to generate one token

$N_{compute} \cdot C \cdot V^2$ switching to compute + $N_{data} \cdot C \cdot V^2$ switching to move data + $N_{idle} \cdot C \cdot V^2$ clock switching while idle + $P_{leak} \cdot T_{token}$ leakage over time

N number of switching events **C** capacitance switched per event (longer wire, larger C) **V** supply voltage T_{token} time to generate the token

same work

Compute

Same math as any accelerator for the same model.

One unified mixed-precision unit packs more compute into the same silicon.

Ndata ↓ C ↓ 1

Shorter trips

Fewer trips to far memory, shorter wires.

Neural Core	Cluster	HBM3E
scratchpad	shared memory	off-chip

reuse here first: N ↓ · short wires: C ↓

▸ Keep the work inside the core, then the cluster

Nidle ↓ Ttoken ↓ 2

Fewer waits

Fewer idle cycles, less time per token.

Next operands ready during compute	▸ Feed the next operands
Every transfer split across many paths	▸ Split every transfer
Small messages in their own lane	▸ Keep small messages out
Hardware sync on a dedicated channel	▸ Two hardware sync modes

Energy model: A. P. Chandrakasan et al., IEEE JSSC 27(4), 1992. Architecture: C.-H. Yu et al., ISSCC 2026, paper 2.2; Rebel100 white paper v.06. ▸ marks the slide that shows each mechanism.

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Cloud-native AI inference

Scalable Rack Solution



Cloud-Native Stack

RBLN Compiler/Runtime

native PyTorch support
distributed inference support

Cloud-Native Toolkit

operator supporting
resource discovery,
scheduling, monitoring

Enterprise AI

Operational Simplicity
Kubernetes-native resource discovery, scheduling, orchestration and monitoring

Datacenter Ready
Designed for scalable workloads with built-in observability and lifecycle management

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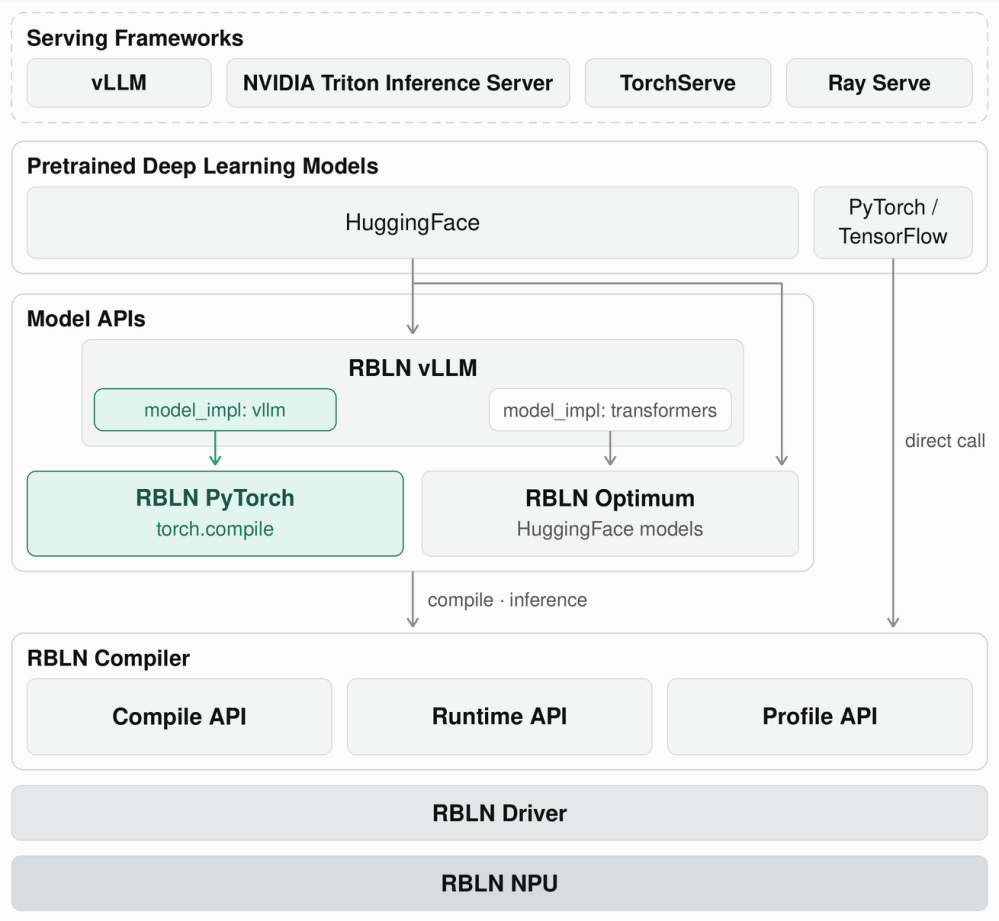
PART 2

Software

From a PyTorch model to rack-scale LLM serving, with less communication overhead and less CPU involvement.

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Keep PyTorch and upstream vLLM — we track for you



Your workflow stays the same

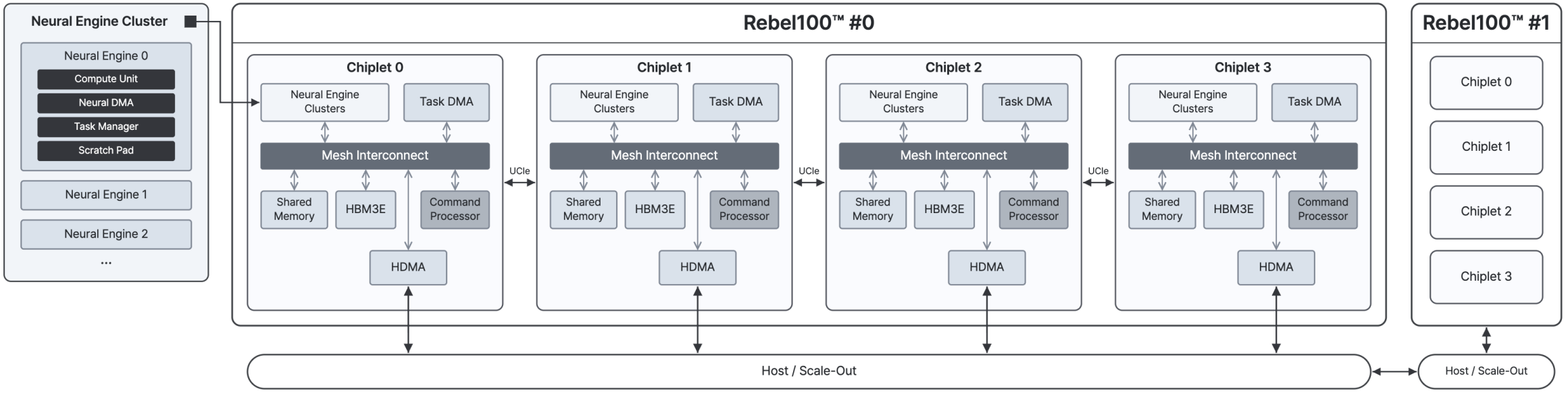
PyTorch	torch.compile with the rbln backend
Hugging Face	models through vllm & optimum-rbln
vLLM	upstream hardware plugin, not a fork
Kubernetes · OpenShift	RBLN NPU Operator

```
$ vllm serve <hugging-face-model-id>
# compiles on first start, then serves from cache
```

Source: docs.rbln.ai, RBLN SDK v0.12.0 (Oct 2026).

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Chiplet to node — one model spans it all



4 chiplets → 1 device

The compiler splits each model across the four chiplets. No user setting.

8 cards → 1 node

Choose vLLM parallelism options. Model code stays the same.

Weights that fit in one node (1,152 GB)

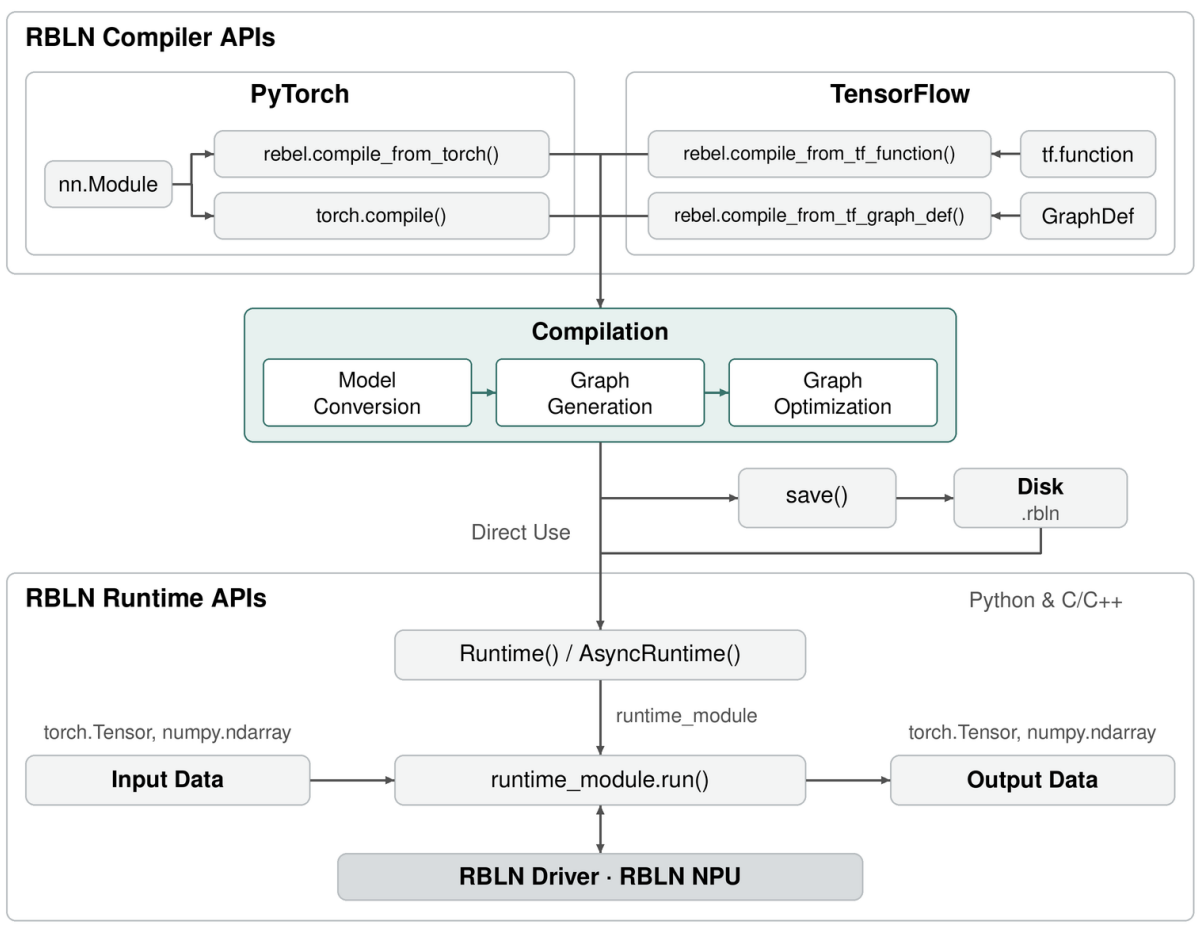
FP16	**~570B**
FP8	**~1.1T**
MXFP4	**~2.2T**

Weights only. KV cache sets the practical limit lower.

Sources: docs.rbln.ai (Model Parallelism, DeepSeek-V3.2 recipe).

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Planned at compile time, enforced on chip



The compiler decides, before anything runs

Whole graph

↓

Partition across 4 chiplets

↓

Schedule DMA and compute together

↓

Command stream

At run time, the runtime fills in device addresses once, then just submits prebuilt command buffers.

Source: docs.rbln.ai, RBLN Compiler overview and Compiler Deep Dive.

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Inside the compiler: nine passes, fusions shipping in the SDK

9 PASSES — RBLN COMPILER WHITE PAPER

1 — Graph opt. + op fusion

2 — Split & group

3 — Tiling

4 — Op scheduling

5 — Bufferization

6 — Memory allocation

7 — Dependency analysis

8 — Command scheduling

9 — Code generation

PASS 1 IN PRACTICE — RELEASE NOTES

GEMM + element-wise

Reshape sunk out → Dense + Add fuse

BEFORE			→	AFTER	
Dense	Reshape	Add	→	Dense + Add	Reshape

Low-precision weights

Dequant runs inside the kernel

BEFORE			→	AFTER	
FP8/MXFP4	Dequant	MatMul	→	FP8/MXFP4	Dequant + MatMul

Rotary embedding

cos/sin transposed once

BEFORE			→	AFTER	
cos/sin	Transpose ×N	RoPE	→	Transpose once	RoPE

Fewer round trips through memory, less redundant work.

Sources: RBLN Compiler white paper; RBLN SDK release notes v0.10.4–v0.12.0.

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The chattiest traffic never leaves the package

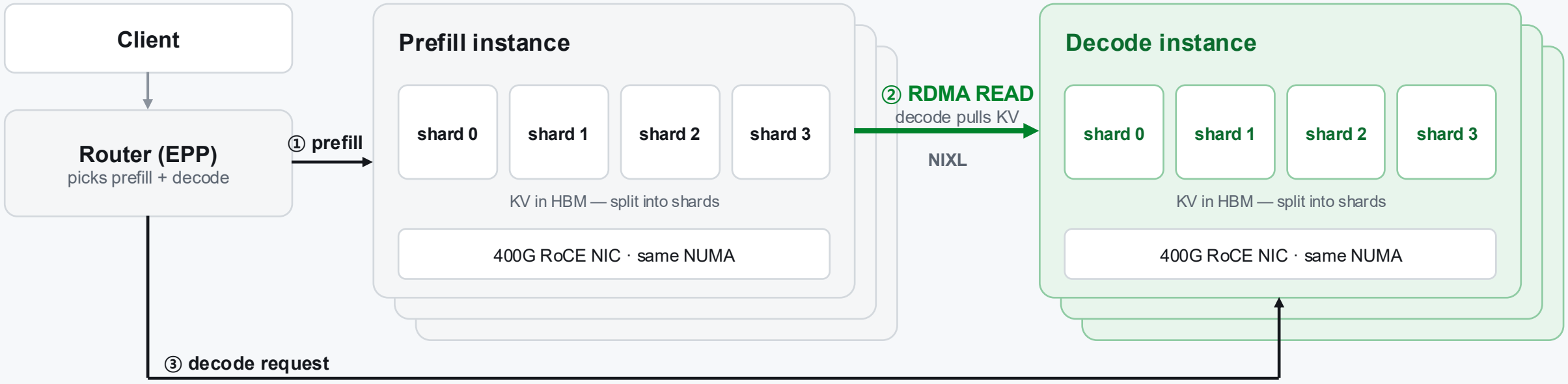
TIER	LINK	BANDWIDTH	WHAT TRAVELS	HANDLED BY
Chiplet ↔ chiplet	UCIe-Advanced	**1 TB/s per channel**	every layer results	`RBLN Compiler`
Card ↔ card	PCIe Gen5 x16 P2P	**~64 GB/s per direction**	collectives traffic	`rbln-ccl`
Node ↔ node	RoCE v2, 400 GbE	**400 GbE/s per NIC**	KV cache (P/D)	`nixl-rbln`

More frequent → closer link. Multi-card serving uses standard vLLM parallelism options.

Sources: docs.rbln.ai (Model Parallelism, rbln-ccl, Disaggregated Prefill/Decode); C.-H. Yu et al., ISSCC 2026.

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Long prompts never stall decode — KV moves NPU to NPU



The NIC reads NPU memory directly: no copy through host memory. One vLLM KV-transfer option turns it on.

Sources: docs.rbln.ai, Disaggregated Prefill/Decode; Rebellions RCNS 2026.10.

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Big blocks, prefix reuse, and TB-scale KV

① BIG BLOCK → BANDWIDTH

one KV block · 4,096 tokens

one large transfer

② SUB-BLOCK PREFIX HITS → FASTER TTFT

Full-block match (upstream)

all recomputed

3,000 to prefill

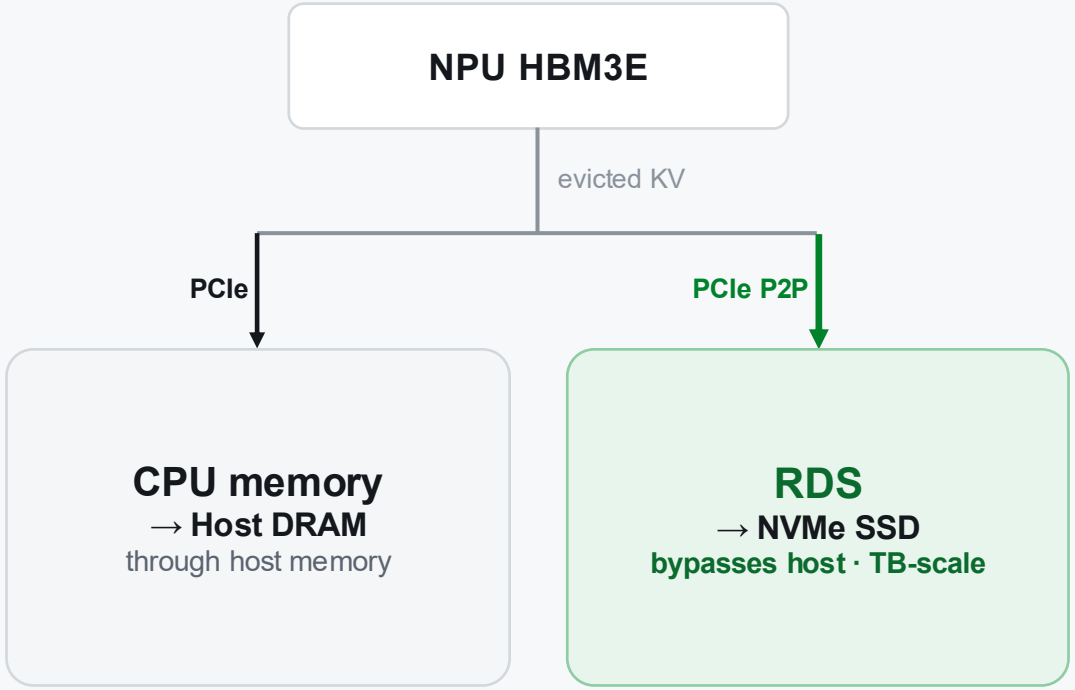
RBLN sub-block match (e.g. 128)



56 to prefill

2,944 reused

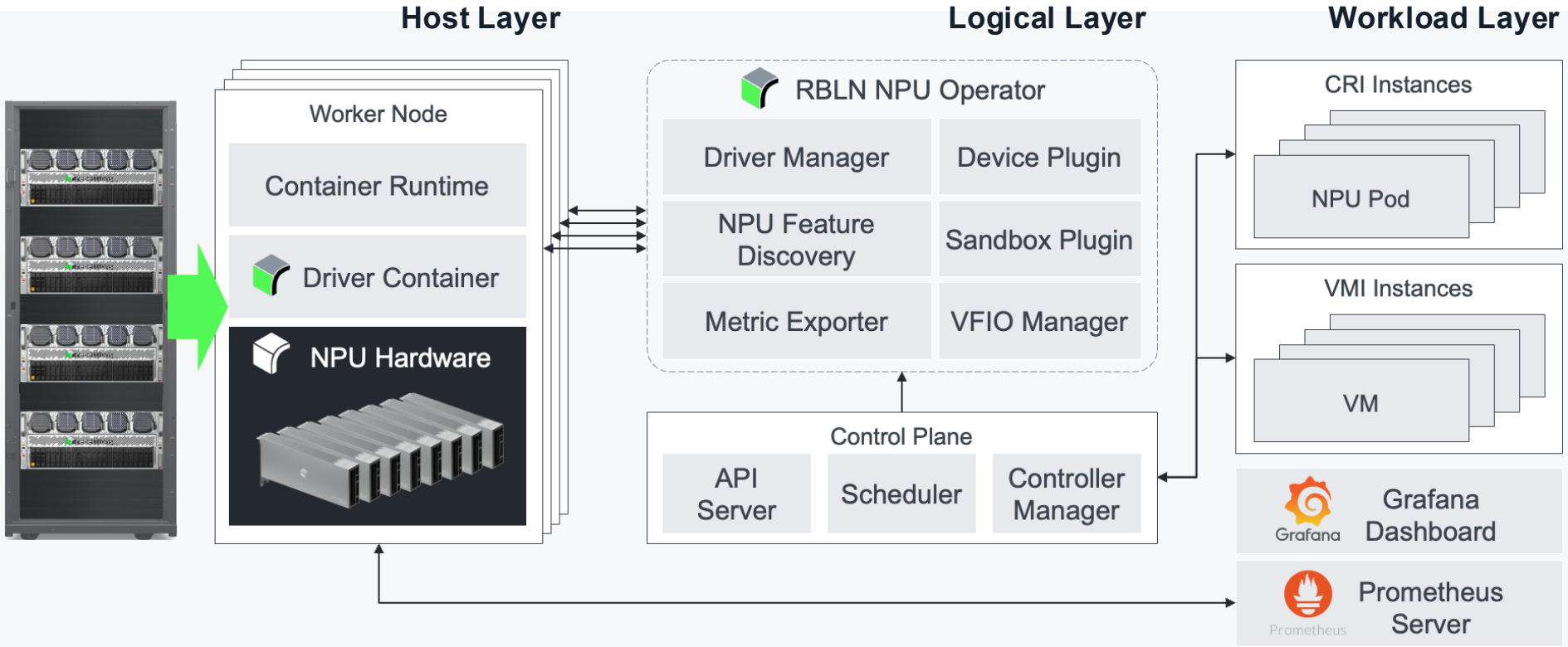
BEYOND HBM — RESTORE, DON'T RECOMPUTE



Sources: docs.rbln.ai, Prefix Caching, KV Cache Offloading, RBLN Direct Storage.

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GPU-style operations — one operator from host to workload



One Helm install

Driver, device plugin or DRA, feature discovery, metrics and VFIO, managed together

NUMA-aware placement

DRA puts the NPU and its NIC on the same NUMA node

Standard monitoring

NPU telemetry into Prometheus and Grafana

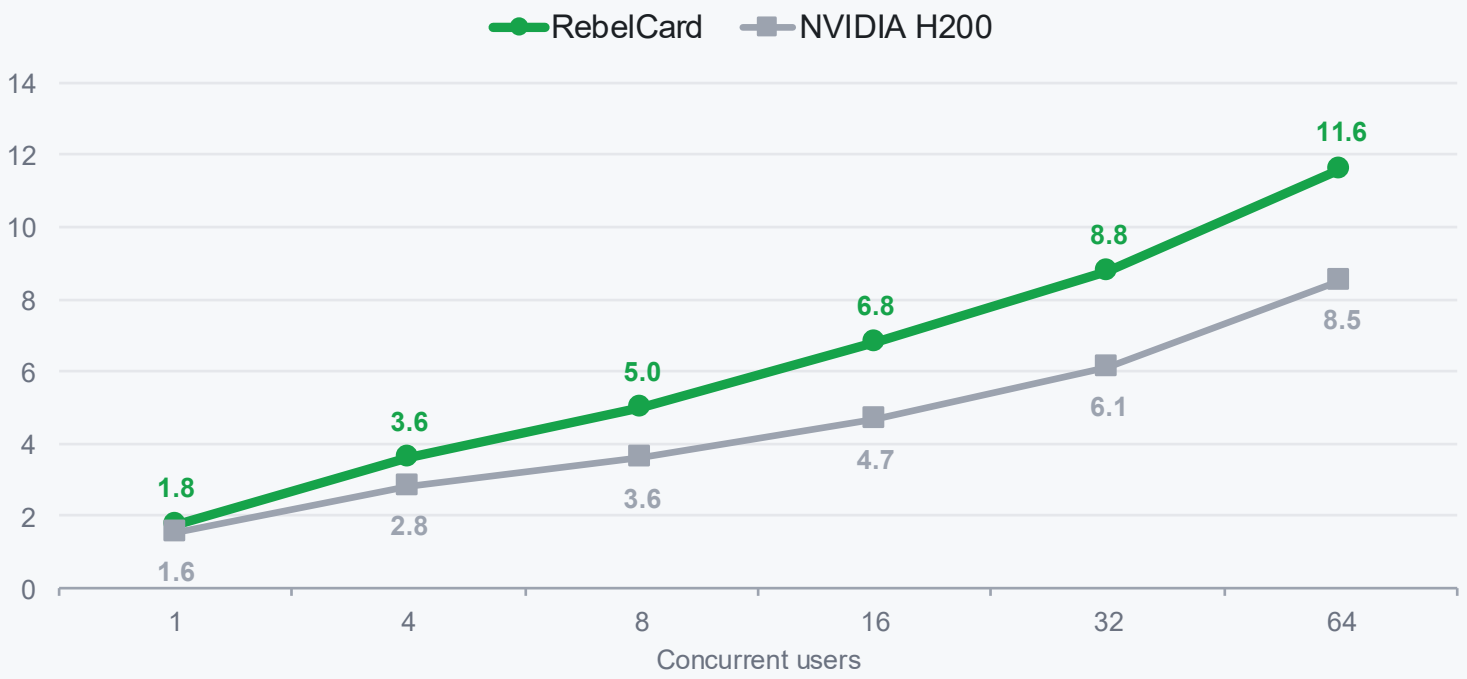
Sources: docs.rbln.ai (NPU Operator, Helm chart, DRA driver); Red Hat Ecosystem Catalog (RBLN NPU Operator).

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More Tokens per Watt Than NVIDIA H200

GPT-OSS-120B · 1 RebelCard vs. 1 H200

Tokens per second per watt higher is better



RebelCard advantage	1.15×	1.27×	1.39×	1.46×	1.43×	1.36×

1.46×
more tokens per watt
peak advantage (16 users)

1.15–1.46×
at every load level

1.95×
tokens per watt per dollar

Test conditions: InferenceX, GPT-OSS-120B; 1× RebelCard vs. 1× NVIDIA H200 (HGX); input / output length 1,024 / 1,024 tokens; TPS divided by mean measured device power. R100: RBLN-CR13, GA SDK 0.12.0. H200: vLLM 0.30.0. Price per watt per dollar uses $300k vs. $400k system price.

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PART 3

How to use RebelServer

Validate recipes, live serving and demos.

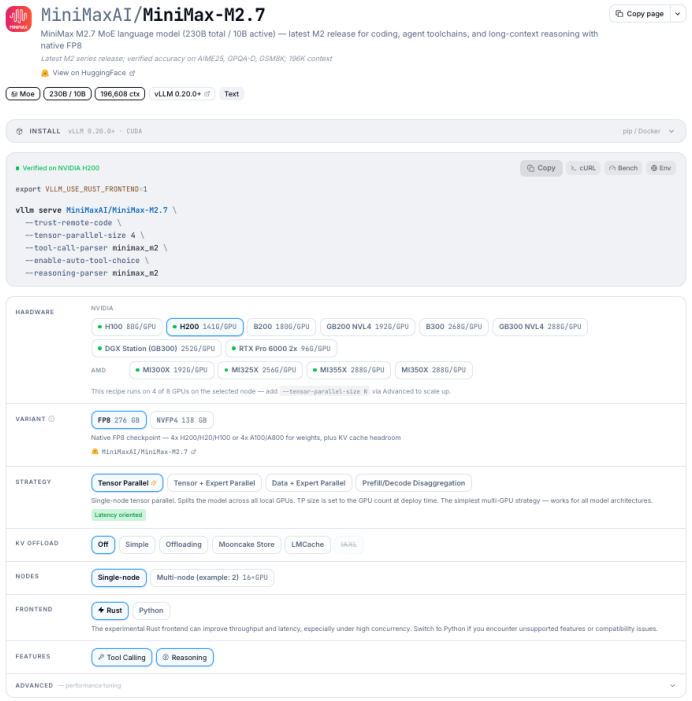


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Start from a validated recipe — it writes the command

vLLM Upstream recipe

https://recipes.vllm.ai/MiniMaxAI/MiniMax-M2.7



rebellions_ RebelCard recipe

docs.rbln.ai/latest/ko/software/model_serving/vllm_support/recipes/minimax-m2-7.html

MiniMaxAI/MiniMax-M2.7

MoE | 230B | 204,800 ctx | vllm-rbln | Text

RBLN-CR13 · 4 NPUs · Flash Attention · ✓ verified 2026-10-07

```
$ LMCACHE_CHUNK_SIZE=4096 LMCACHE_LOCAL_CPU=False LMCACHE_MAX_LOCAL_CPU_SIZE=0 \
LMCACHE_EXTRA_CONFIG='{"storage_plugin.RDSBackend.module_path": "lmcache_rbln.storage_plugin.rds_backend",
  "storage_plugin.RDSBackend.class_name": "RDSBackend"}' \
LMCACHE_STORAGE_PLUGINS=RDSBackend PYTHONHASHSEED=42 VLLM_ENGINE_READY_TIMEOUT_S=9600 \
vllm serve MiniMaxAI/MiniMax-M2.7 \
    --block-size 4096 --data-parallel-size 4 \
    --enable-auto-tool-choice --enable-expert-parallel \
    --gpu-memory-utilization 0.8 --kv-cache-dtype fp8 \
    --kv-transfer-config '{"kv_connector": "RBLNLMCacheConnectorV1", "kv_role": "kv_both"}' \
    --max-model-len 204800 --max-num-batched-tokens 512 --max-num-seqs 1 \
    --model-impl vllm --prefix-caching-hash-algo sha256_cbor \
    --reasoning-parser minimax_m2 --safetensors-load-strategy prefetch \
    --speculative-config '{"method": "eagle3", "model": "asherszhang/MiniMax-M2.7-EAGLE3-draft-vocab200k",
      "num_speculative_tokens": 3}' \
    --tool-call-parser minimax_m2 --trust-remote-code
```

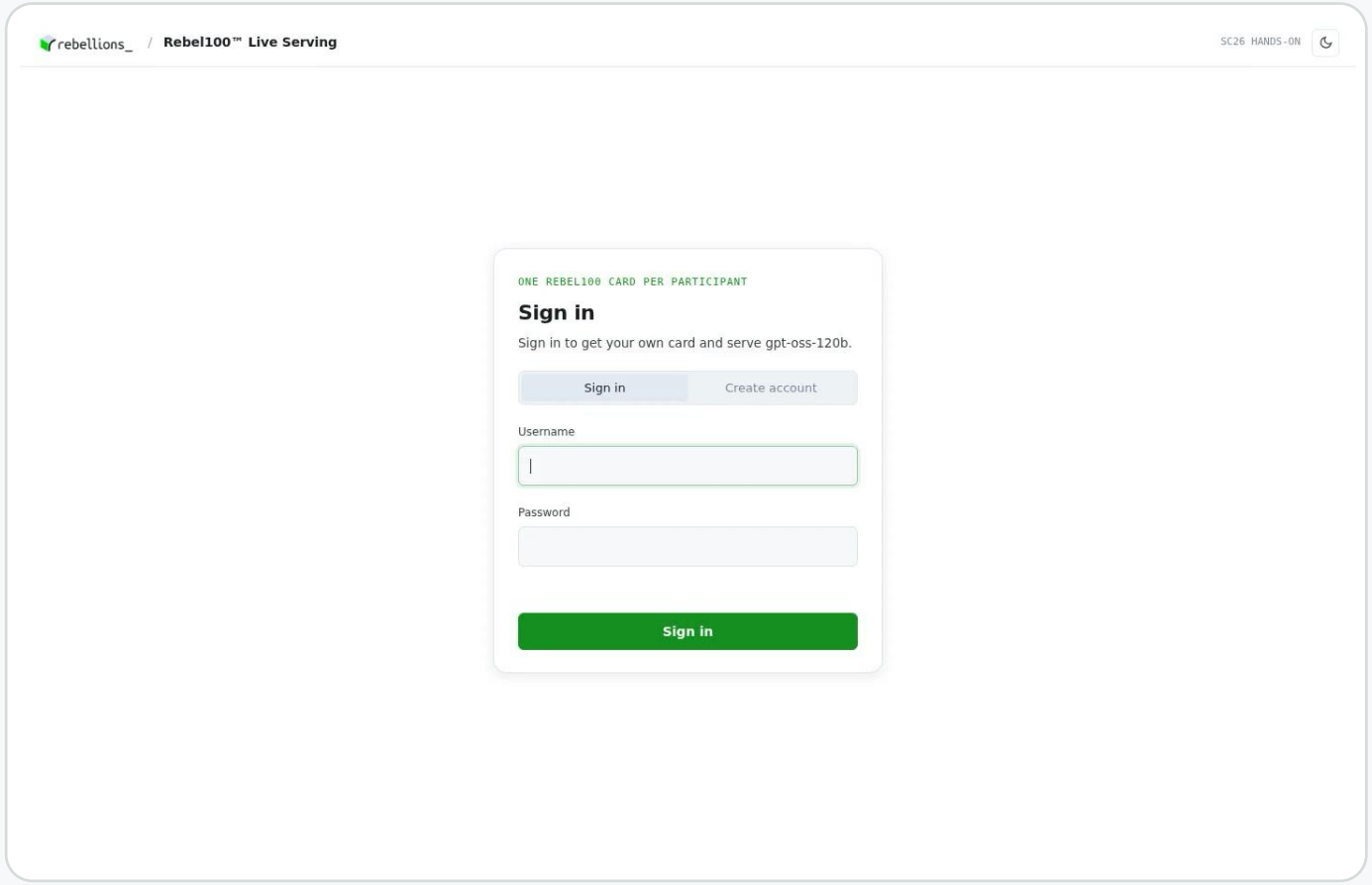
Hardware	**RBLN-CR13 ×4**	Variant	**FP8**
Model impl	**vllm**	Node	**Single node**
Strategy	**Aggregated · Data + Expert Parallel** / Aggregated · Pipeline Parallel / Prefill/Decode disaggregated		
Optimize for	**Latency** / Throughput	KV offload	Off / LMCache (CPU) / **LMCache (RDS)**
Prefix caching	On / **On (sub-block)** / Off	Spec. decoding	Off / **EAGLE-3** / Suffix
Reasoning parser	Off / **On**	Tool calling	Off / **On**

Source: docs.rbln.ai, MiniMax-M2.7 recipe (verified 2026-10-07). Command shortened for display.

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Try it live — Rebel100 serving, right now

gpt-oss-120b on one RBLN-CR13



Live chat | Card power + NPU memory | Recipe options → serve command | Serve log

SCAN TO GET DEMO LINK

SECRET CODE
sc26-demo

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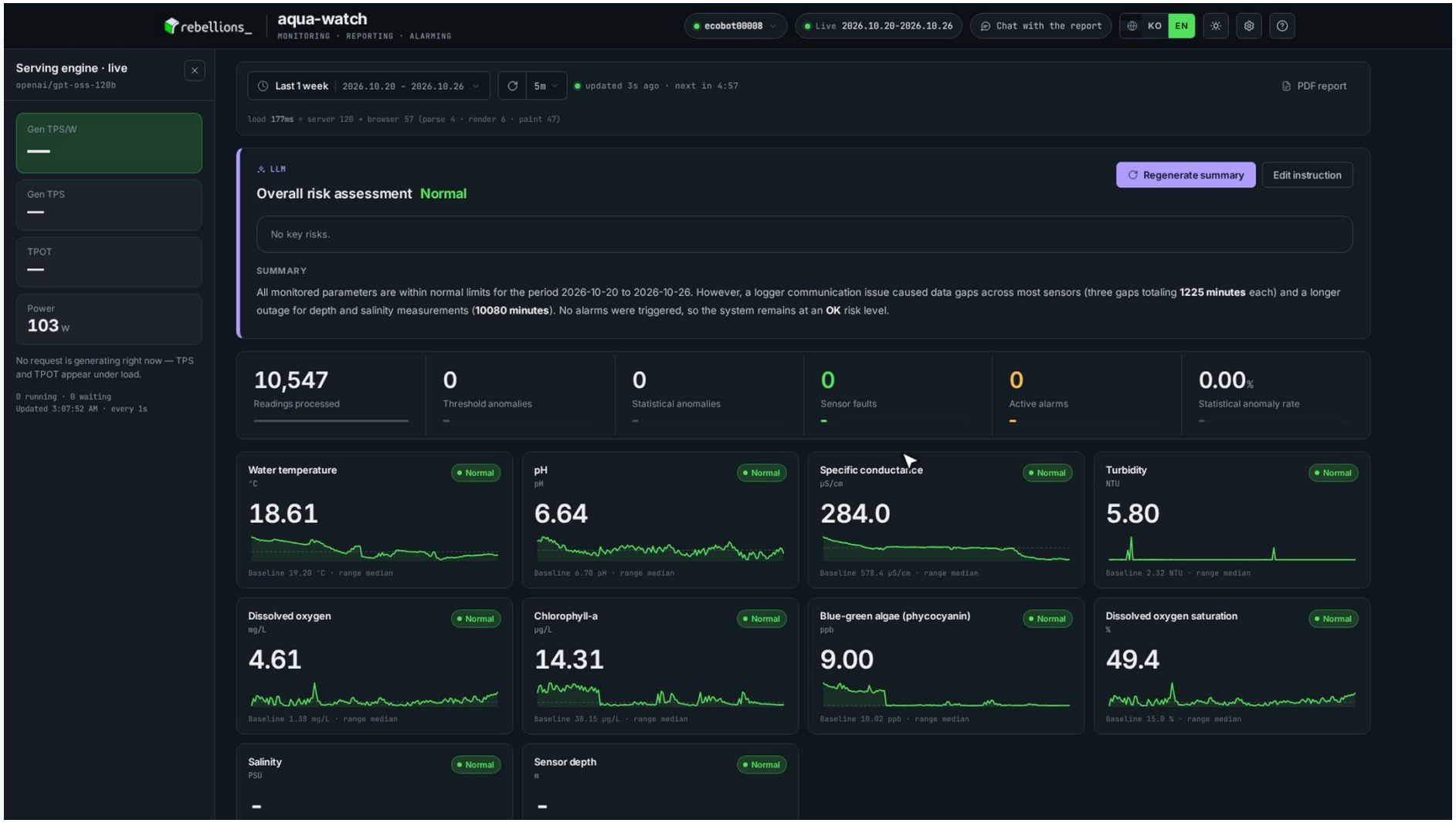
Demo: Aqua Report

VIDEO DEMO

See it at our booth

Get the full Aqua Report walkthrough from our team.

Meet Rebellions at booth #7003



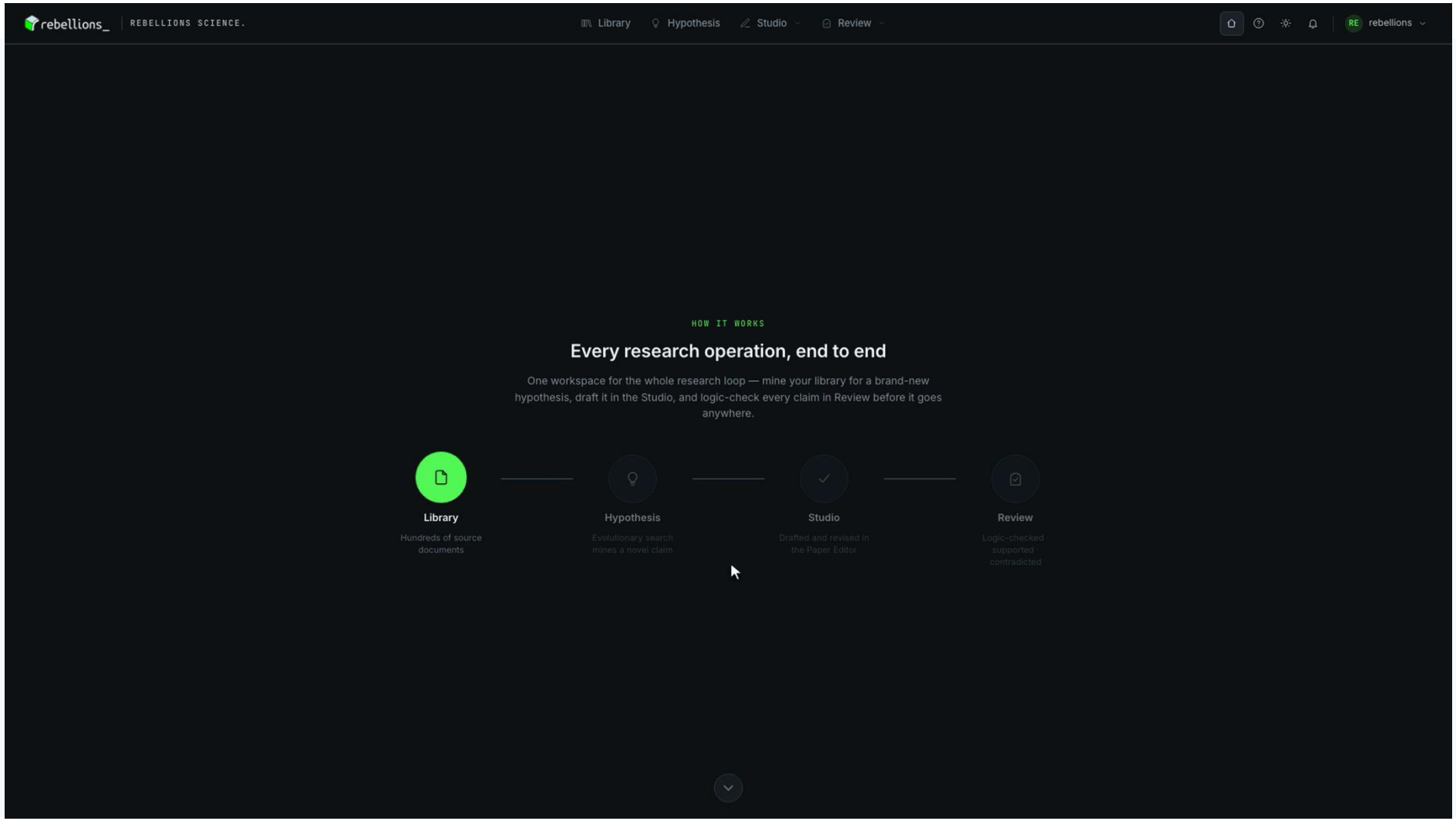
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Demo: Rebellions Science

LIVE DEMO

Try it live at our booth

Run Rebellions Science yourself, from library to paper draft.



Meet Rebellions at booth #7003

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Thank You

Keep data close. Keep it flowing.

Keep execution in step.

docs.rbln.ai

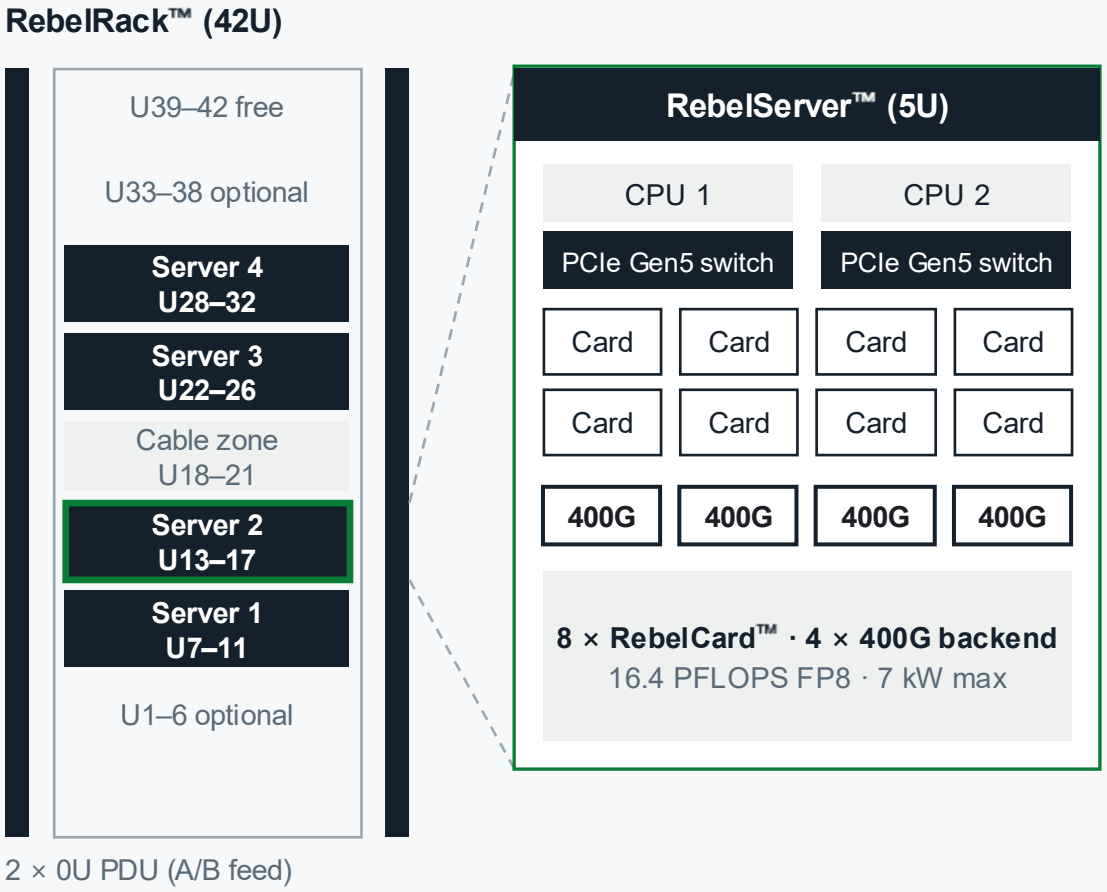
SCAN FOR LINKS

Docs, white paper, publications, GitHub



From server to rack

8 cards per server, 4 servers per rack



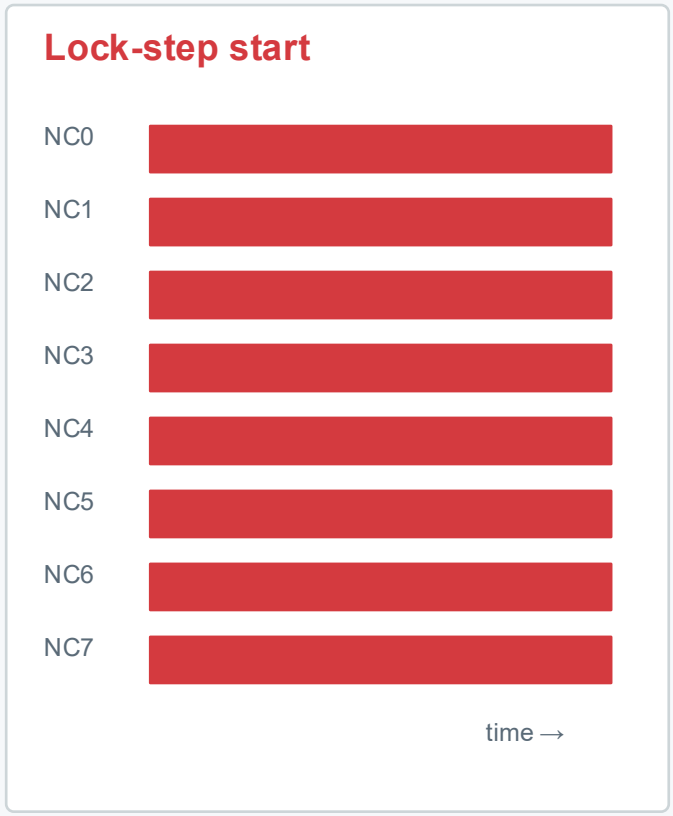
	RebelServer™	RebelRack™
Configuration	5U, 8 × RebelCard™	42U, 4 servers (32 cards)
FP8 / FP16	16.4 / 8.2 PFLOPS	65.5 / 32.7 PFLOPS
HBM3E	1,152 GB · 38.4 TB/s	4.6 TB · 153.6 TB/s
Host	2 × EPYC 9355, 1.5 TB	8 × EPYC 9355, 6 TB
Storage	4 × 1.92 TB U.2 NVMe	16 × 1.92 TB U.2 NVMe
Compute fabric	4 × 400G per server	800G switches
Front-end	2 × 100G, 2 × 25G (OCP 3.0)	25G switch, 1GbE OOB
Power	7 kW max (4–6 typical)	28 kW max (16–24 typical)
Cooling	Air	Air, front-to-back
Software	RHEL / Ubuntu + Rebel SW	Same

Sources: RebelServer™ datasheet v1.6; RebelRack™ standard spec.

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Stagger the start so the chip can hold its peak

3 · KEEP EXECUTION IN STEP





0.74 → 0.22

A/ns current ramp per chiplet, lock-step vs. staggered

> 20 mVpp

supply noise suppressed

Same work, spread over time: a gentler ramp keeps the supply steady and the peak reachable.

Source: C.-H. Yu et al., ISSCC 2026, paper 2.2 (Fig. 2.2.4); schematic redrawn.

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